

26 Central Limit Theorem

- Many statistical applications involve *random sampling* from a population.
- And one of the most basic operations in statistics is to average the values in a sample.
- If a sample is selected at random, then the sample mean is a random variable that has a distribution. How does this distribution behave?

Example 26.1. A random sample of n customers at a snack bar is selected, independently. Let T_n be the total dollar amount spent by the n customers in the sample, and let $\bar{X}_n = T_n/n$ represent the sample mean dollar amount spent by the n customers in the sample.

Assume that

- 40% of customers spend 5 dollars
- 40% of customers spend 6 dollars
- 20% of customers spend 7 dollars

Let X denote the amount spent by a single randomly selected customer.

1. Find $E(X)$. We'll call this value the population mean μ ; explain what this means.
2. Find $SD(X)$. We'll call this value the population standard deviation σ ; explain what this means.
3. Randomly select two customers, independently, and let X_1 and X_2 denote the amounts spent by the two people selected. Make a table of all possible (X_1, X_2) pairs and their probabilities.

4. Use the table from the previous part to find the distributions of T_2 and $\bar{X}_2$. Interpret in words in context what these distributions represent.
5. Compute and interpret $P(\bar{X}_2 > 6)$.
6. Compute $E(\bar{X}_2)$. How does it relate to μ ?
7. There are 3 “means” in the previous part. What do they all mean?
8. Compute $\text{Var}(\bar{T}_2)$ and $\text{Var}(\bar{X}_2)$. How do these values relate to the population variance σ^2 ?
9. Compute $\text{SD}(\bar{T}_2)$ and $\text{SD}(\bar{X}_2)$. How do these values relate to the population standard deviation σ ?
10. Describe in words in context what $\text{SD}(\bar{X}_2)$ measures variability of.

- The **population distribution** describes the distribution of values of a variable over all *individuals* in the population.
 - The **population mean**, denoted μ , is the average of the values of the variable over all individuals in the population.
 - The **population standard deviation**, denoted σ , is the standard deviation of all the individual values in the population.
- A **(simple) random sample** of size n is a collection of random variables $X_1, \dots, X_n$ that are *independent* and *identically distributed (i.i.d.)*
- The **sample mean** is

$$\bar{X}_n = \frac{X_1 + \dots + X_n}{n} = \frac{T_n}{n},$$

- Because the sample is randomly selected, the sample mean $\bar{X}_n$ is a random variable that has a distribution. This distribution describes how sample means vary from sample-to-sample over many random samples of size n .
- Over many random samples, sample means do not systematically overestimate or underestimate the population mean.

$$E(\bar{X}_n) = \mu$$

- Variability of sample means depends on the variability of individual values of the variable; the more the values of the variable vary from individual-to-individual in the population, the more the sample means will vary from sample-to-sample. However, sample means are less variable than individual values of the variable. Furthermore, the sample-to-sample variability of sample means decreases as the sample size increases

$$\begin{aligned} \text{Var}(\bar{X}_n) &= \frac{\sigma^2}{n} \\ \text{SD}(\bar{X}_n) &= \frac{\sigma}{\sqrt{n}} \end{aligned}$$

Example 26.2. Continuing Example 26.1. Now suppose we take a random sample of 30 customers; consider $n = 30$ and $\bar{X}_{30}$.

1. Find $E(\bar{X}_{30})$, $\text{Var}(\bar{X}_{30})$, and $\text{SD}(\bar{X}_{30})$.

2. What does $\text{SD}(\bar{X}_{30})$ measure the variability of?
 3. Use simulation to determine the approximate shape of the distribution of $\bar{X}_{30}$.
 4. Simulation shows that $\bar{X}_{30}$ has an approximate Normal distribution. Use this Normal distribution to approximate $P(\bar{X}_{30} > 6)$, and interpret the probability.
- **The (Standard) Central Limit Theorem.** Let $X_1, X_2, \dots$ be independent and identically distributed (i.i.d.) random variables with finite mean μ and finite standard deviation σ . Then

$\bar{X}_n$ has an approximate $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ distribution, if n is large enough.

T_n has an approximate $N(n\mu, \sigma\sqrt{n})$ distribution, if n is large enough

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means (or sums) is approximately Normal, *regardless of the shape of the population distribution*.

Example 26.3. Continuing Example 26.1. Now suppose that each customer spends 5 dollars with probability 0.4, 6 with probability 0.4, 7 with probability 0.19, and 30 with probability 0.01 (maybe they treat a few friends).

1. Use simulation to approximate the distribution of $\bar{X}_{30}$; is it approximately Normal?

2. Use simulation to approximate the distribution of $\bar{X}_{100}$; is it approximately Normal?

3. Use simulation to approximate the distribution of $\bar{X}_{300}$; is it approximately Normal?

- The CLT says that if the sample size is large enough, the sample-to-sample distribution of sample means is approximately Normal, *regardless of the shape of the population distribution*.
- However, the shape of the population distribution — which describes the distribution of individual values of the variable — does matter in determining *how large* is large enough.
 - If the population distribution is Normal, then the sample-to-sample distribution of sample means is Normal for any sample size.
 - If the population distribution is “close to Normal” — e.g., symmetric, light tails, single peak — then smaller samples sizes are sufficient for the sample-to-sample distribution of sample means to be approximately Normal.
 - If the population distribution is “far from Normal” — e.g., severe skewness, heavy tails, extreme outliers, multiple peaks — then larger sample sizes are required for the sample-to-sample distribution of sample means to be approximately Normal.
- You should certainly be aware that there is no magic number and that some population distributions require a large sample size for the CLT to kick in. However, in many situations the sample-to-sample distributions of sample means (or other statistics) is approximately Normal even for small or moderate sample sizes.

Example 26.4. The number of free throws that a particular basketball player attempts in a game has a Poisson distribution with parameter 2.7, independently from game to game. Approximate the probability that the player attempts more than 60 free throws in the next 20 games.

Example 26.5. Ten independent resistors are connected in series. Each has a resistance Uniformly distributed between 215 and 225 ohms, independently. Compute the probability that the mean resistance of this series system is between 219 and 221 ohms.

Example 26.6. A fair six-sided die is rolled until the total sum of all rolls exceeds 300. Approximate the probability that at least 80 rolls are necessary.

Example 26.7. Suppose that average annual income for U.S. households is about \$100,000, and that the standard deviation of income is about \$230,000.

1. What can you say about the probability that a single randomly selected household has an income above \$120,000?

2. Donny Don't says: "Since a sample size of one million is extremely large, the CLT says that the incomes in a sample of one million households should follow a Normal distribution". Do you agree? If not, explain to Donny what the CLT does say.

3. Use the CLT to approximate the probability that in a sample of 1000 households the sample mean income is above \$120,000.

4. Donny says: “Wait, the distribution of household income is highly skewed to the right. Wouldn’t it be more appropriate to use the sample median than the sample mean”? Do you agree? Explain.